Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/15/2013
PI Name:	Turner, Russell T Ph.D.		
Project Title:	Role of Marrow Adipocytes in Bone Loss during Simulated Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Corvallis	**State:**	OR
Zip Code:	97331-8558	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	09/15/2012	**End Date:**	09/14/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	There were no changes in key personnel.		
COI Name (Institution):	Iwaniec, Urszula (Oregon State University) Wong, Carmen (Oregon State University) Zhang, Bo (Oregon State University)		
Grant/Contract No.:	NNX12AL24G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight, by impairing the differentiation program of stem cells residing within bone marrow results in bone loss, increased bone marrow adiposity, anemia and impaired immune function. Because these disturbances may potentially compromise the success of long-duration flights, there exists an urgent need to identify the underlying mechanisms and implement effective countermeasures. We propose to test the novel hypothesis that negative energy balance during spaceflight results in 1) bone marrow dysfunction and increased marrow adiposity and 2) altered signaling by peripheral adipokines (cytokines produced by adipocytes) all of which, in turn, contribute to negative bone turnover balance and bone loss. The proposed research will focus on the precise role of adipose tissue and the important adipokine leptin in mediating the detrimental effects of simulated spaceflight (hindlimb unloading) on bone marrow cells in mice by accomplishing the following 2 specific aims: Specific Aim 1 will test the hypothesis that increased bone marrow adiposity plays a causal role in mediating simulated spaceflight-induced changes in bone metabolism. This will be accomplished by comparing the skeletal response to hindlimb unloading of wild type (WT) mice with mice incapable of generating bone marrow adipocytes (kit-receptor-deficient KitW/W-v mice). Specific Aim 2 will test the hypothesis that the adipokine leptin, by targeting the differentiation of bone marrow stem cells, plays a key role in mediating energy sensitive changes in bone metabolism during simulated spaceflight. This will be accomplished by examining whether transplantation of mesenchymal or hematopoietic stem cells from leptin receptor-deficient db/db mice into WT mice alters bone loss during simulated spaceflight. Completion of the proposed project is expected to increase our understanding of the role of energy metabolism in mediating bone loss during simulated spaceflight and suggest specific non-pharmacological and pharmacological interventions to mitigate the detrimental effects of spaceflight on bone marrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a major health concern. To our knowledge, all forms of osteoporosis involve infiltration of fat into bone marrow. If the increased marrow fat contributes to bone loss, as hypothesized, this research will have significance to human health that transcends protecting astronauts from bone loss during spaceflight.
Task Progress:	During spaceflight and simulated spaceflight (extended bed rest in humans and hind limb unloading in rodents) bone marrow becomes infiltrated with adipose tissue. Specific Aim 1 of this proposal is designed to test the hypothesis that increased bone marrow fat plays a causal role in mediating simulated spaceflight-induced detrimental changes in bone metabolism. The changes in bone metabolism include increased bone resorption and decreased bone formation. The net result is rapid bone loss. The first step of the proposal was to build and test cages for performing hind limb unloading of mice. We have accomplished this task. Additionally, students working on the project were trained to handle experimental animals and perform the skeletal analyses that will be used in this research. We have also initiated the first major experiment of the project. This 4-month-long study involved comparing the skeletal response to hind limb unloading of mice with a mutation preventing production of bone marrow fat with normal mice. If our hypothesis is correct than the mutant mice will exhibit less bone loss during simulated spaceflight then normal mice. We have completed the animal portion of experiment and are in the process of analyzing the skeleton by a variety of high resolution x-ray based scanning techniques and histology.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Iwaniec UT, Turner RT. "Failure to generate bone marrow adipocytes does not protect mice from ovariectomy-induced osteopenia." Bone. 2013 Mar;53(1):145-53. Epub 2012 Dec 12. http://dx.doi.org/10.1016/j.bone.2012.11.034 ; PubMed PMID: 23246792 , Mar-2013
Articles in Peer-reviewed Journals	Turner RT, Iwaniec UT, Wong CP, Lindenmaier LB, Wagner LA, Branscum AJ, Menn SA, Taylor J, Zhang Y, Wu H, Sibonga JD. "Acute exposure to high dose ?-radiation results in transient activation of bone lining cells." Bone, Available online 2013 Aug 13. http://dx.doi.org/10.1016/j.bone.2013.08.002 , Aug-2013